

NEXUS AI: A Multi-Agent Agentic AI System for Software-Defined Vehicle Engineering, Simulation, and Validation

Team Members

KOWSHIKA I R (231501080)

KRISHNA KUMAR N K (231501081)

LAKSHIYA SRI K M (231501082)



Problem Definition

- SDV engineering involves multiple interconnected activities. Current workflows are distributed across different tools and processes.
- Manual coordination is required between research, design, coding, safety review, and validation.
- Conventional LLMs mainly generate answers or code without complete engineering validation.
- Weak traceability exists between Requirement → Design → Code → Validation.
- Simulation failures may require repeated manual analysis and correction.

Business / Real-World Context

- Automotive OEMs – SDV development and engineering workflows
- Tier-1 Suppliers – Component and software development
- Validation Teams – Scenario testing and telemetry analysis
- SDV Software Teams – Rapidly changing software requirements
- Engineering Managers – Monitoring progress, risks, decisions, and approvals

Target Users

- Automotive Software Engineers
- System Architects
- Safety Engineers
- Validation Engineers
- Engineering Managers
- Human Reviewers

Stakeholder Identification

Stakeholder	Main Need
Software Engineer	Generate and validate implementations
System Architect	Design architecture and signal mapping
Safety Engineer	Identify risks and edge cases
Validation Engineer	Run simulations and analyze telemetry

3 **Proposed System: Analysis Pipeline**

Transforms a natural-language automotive requirement into a validated and traceable engineering solution through a multi-agent workflow.

1 Requirement Input

Accepts a high-level SDV requirement and initiates the engineering workflow.

2 ORION Orchestration

Breaks the requirement into tasks and coordinates the specialized agents.

3 ATLAS + NOVA

Performs automotive research and generates the system architecture.

4 FORGE + SENTINEL.

Generates implementation code and performs safety/edge-case review

5 PILOT Simulation

Runs CARLA-based simulation, fault injection, and telemetry analysis.

6 Reflection + HITL

Handles failures through correction and requests human approval when required.

Provides automotive knowledge and simulation resources required for research, architecture, implementation, and validation.

COVESA VSS

Vehicle Signal Specification for standardized vehicle data models.

COVESA VISS

Vehicle Information Service Specification for accessing vehicle data.

Automotive Documentation

Curated technical sources used as the knowledge base for Agentic RAG.

Qdrant Vector Database

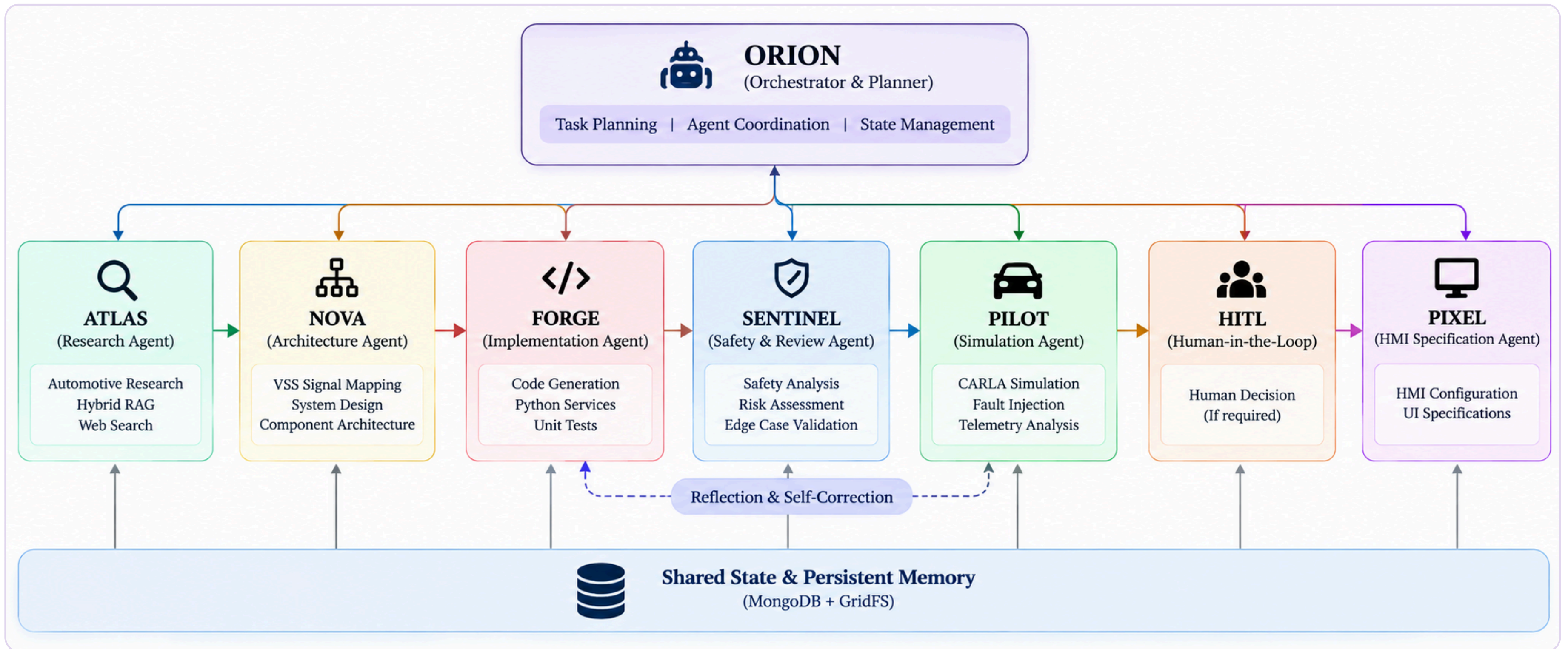
Stores document embeddings for semantic retrieval.

CARLA + EV Physics Model

Provides simulation and telemetry data for validation and fault testing.

5 TOOLS AND TECHNOLOGIES USED

Layer	Technologies / Tools			
1 User Layer		React (Web UI)		
2 Orchestration Layer		LangGraph		Python
3 Agent Layer		LLM (Agents)		Agent Tools
4 Knowledge Layer		Qdrant		BM25
				DuckDuckGo
				Automotive Standards
5 Data Layer		MongoDB		GridFS
6 Infrastructure Layer		Docker		Local / Cloud



The proposed system consists of seven specialized agents coordinated by LangGraph.

ORION — Orchestrator

Role: Project manager

Responsibilities:

- requirement decomposition
- task planning
- agent selection
- global state management

ATLAS — Research Agent

Role: Automotive researcher

Uses:

- Hybrid RAG
- Qdrant
- BM25
- reranking
- web research

NOVA — Architecture Agent

Creates:

- system architecture
- signal mapping
- APIs
- component relationships
- design alternatives

FORGE — Implementation Agent

Generates:

- Python implementation
- CAN decoder
- battery health service
- anomaly detection
- tests

SENTINEL — Safety Agent

Performs:

- hazard analysis
- edge-case analysis
- fail-safe review
- engineering-oriented safety review

PILOT — Simulation Agent

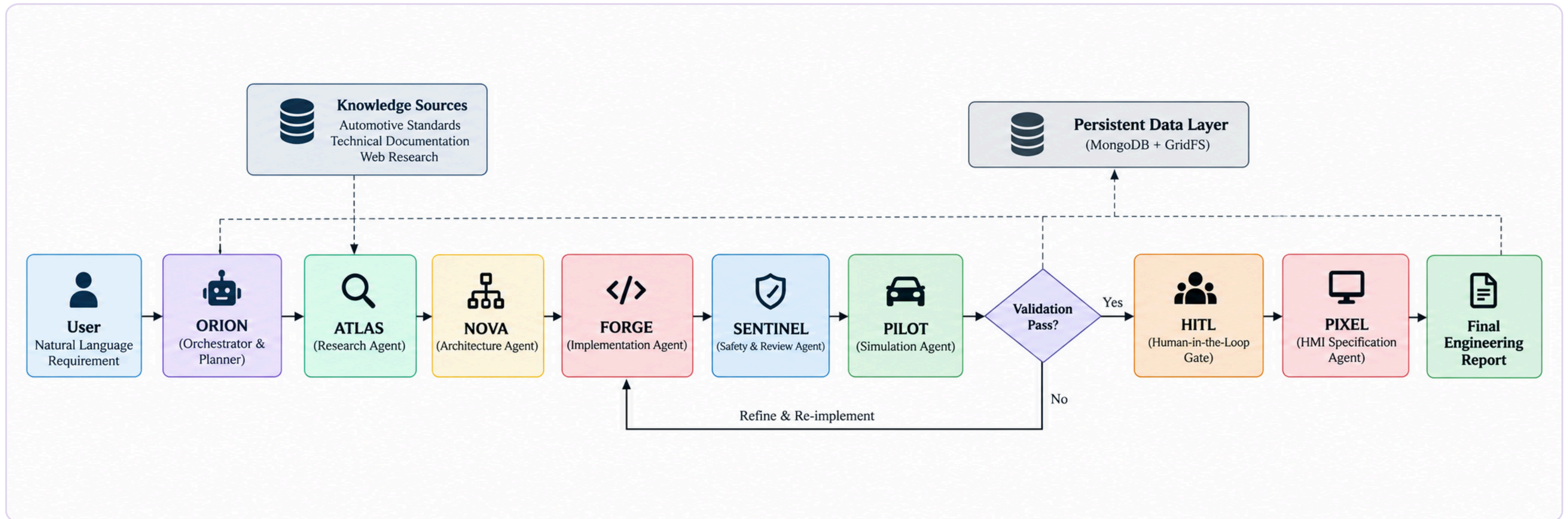
Controls:

- CARLA
- EV physics model
- thermal fault injection
- telemetry
- simulation analysis

PIXEL — HMI Agent

containing:

- vehicle health indicators
- temperature
- SOC
- voltage
- warnings



1

Knowledge Sources

1. User — Natural Language
2. ORION — Orchestrator &
3. ATLAS — Research Agent
4. NOVA — Architecture Agent
5. FORGE — Implementation
6. SENTINEL — Safety & Review Agent
7. PILOT — Simulation Agent
8. Validation & Reflection
9. HITL — Human-in-the-Loop
10. PIXEL — HMI Specification
11. Final Engineering Report

2

Knowledge Sources

Knowledge Sources

- Automotive Standards
- Technical Documentation
- Web Research

“LangGraph is used as the orchestration backbone of NEXUS AI, enabling specialized agents to share state, execute conditionally, perform reflection and self-correction, retry failed tasks, and involve human intervention when required.”

LangGraph

LangGraph acts as the workflow engine, while it acts as the central workflow controller.

3

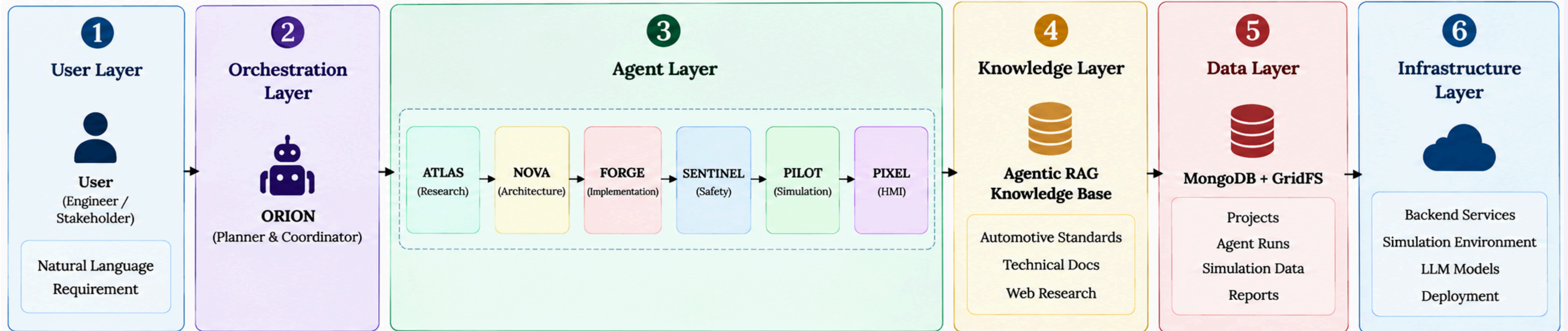
Persistent Data Layer

MongoDB + GridFS

- Stores project information
- Agent execution traces
- Research sources
- Generated artifacts
- Safety reviews
- Simulation data and telemetry
- HITL decisions
- Engineering reports

ORION

ORION acts as the decision-making orchestrator inside that workflow. Maintains the shared workflow state between agents.



1 Knowledge Sources

Automotive Knowledge Base

The knowledge layer supports automotive engineering research through:

Curated Sources

- COVESA VSS
- COVESA VISS
- CARLA documentation

External Research

- DuckDuckGo web research

2 Why Hybrid Retrieval?

Hybrid Retrieval Strategy

NEXUS combines two complementary retrieval approaches:

Dense Retrieval — Qdrant Captures:

Semantic similarity
Useful when the query and source use different wording but describe the same concept.

Sparse Retrieval — BM25 Captures: Exact lexical relevance

3 Retrieval Quality Evaluation

RAG Evaluation

The implementation includes a benchmark comparing:

Baseline 1 — LLM Only

→ No retrieval

Baseline 2 — Basic RAG

→ Retrieval without the complete agentic engineering workflow

Proposed — NEXUS AI

→ Agentic workflow + hybrid RAG + validation + simulation

Main Technologies



FastAPI
Backend API



LangGraph
Agent orchestration



MongoDB
Stores project and
agent information



Qdrant + BM25
Knowledge retrieval



CARLA
CARLA + EV Model
Simulation and testing

Implementation Flow

User Requirement



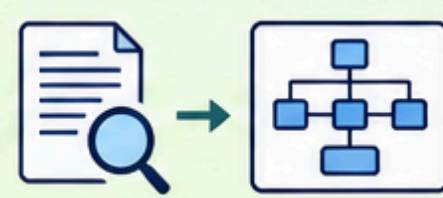
Provide an
automotive requirement

ORION Agent



Understands and
decomposes the
requirement

Research → Architecture



- Research relevant information
- Generate system architecture

Implementation



- Generate code
- Create engineering artifacts
- Integrate components

Safety Review



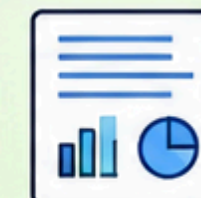
- Analyze safety aspects
- Validate with standards
- Provide feedback

Simulation



- Test in CARLA
- Validate performance
- Evaluate safety metrics

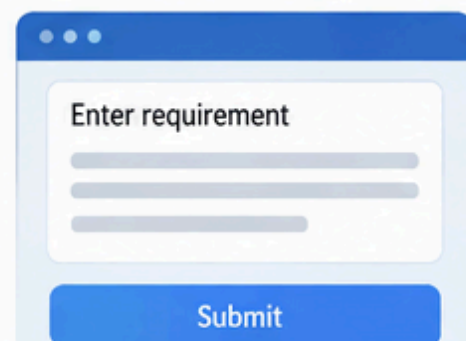
Final Report



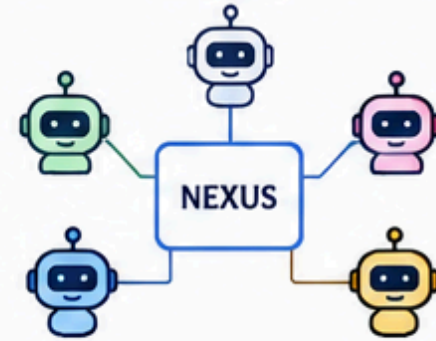
- Generate report
- Summarize results
- Store for traceability

Functional Prototype

1 Accepts an automotive requirement



2 Coordinates multiple AI agents



3 Generates engineering artifacts



4 Performs simulation-based validation



5 Stores the complete workflow for traceability

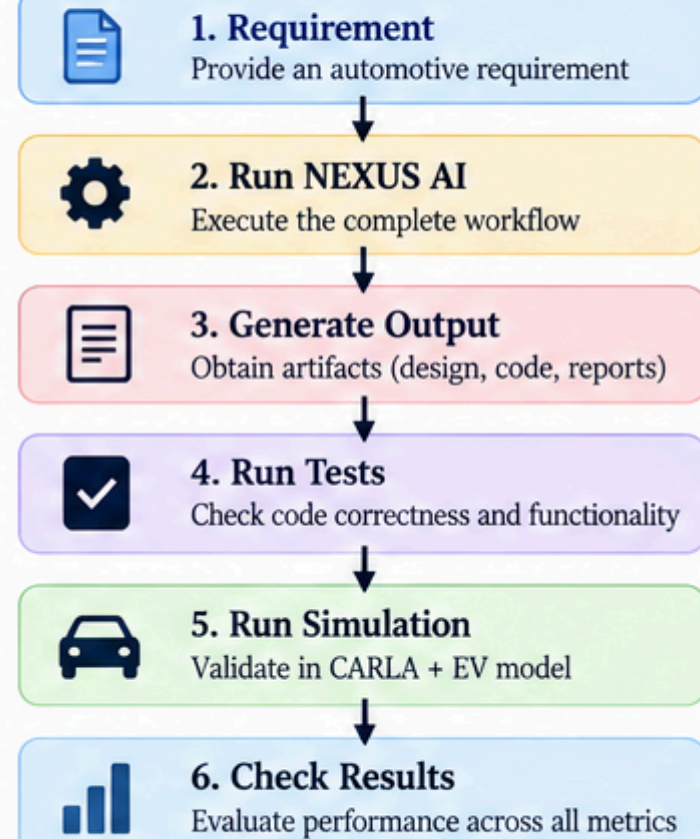


We evaluate NEXUS AI at different stages of the workflow.

Evaluation Areas

Evaluation Area	What We Check
 RAG Quality	Is the retrieved information relevant?
 Citation Quality	Are the sources correct?
 Code Quality	Does the generated code pass tests?
 Simulation	Can the system detect faults?
 Workflow	Can the complete process finish successfully?

Testing Methodology

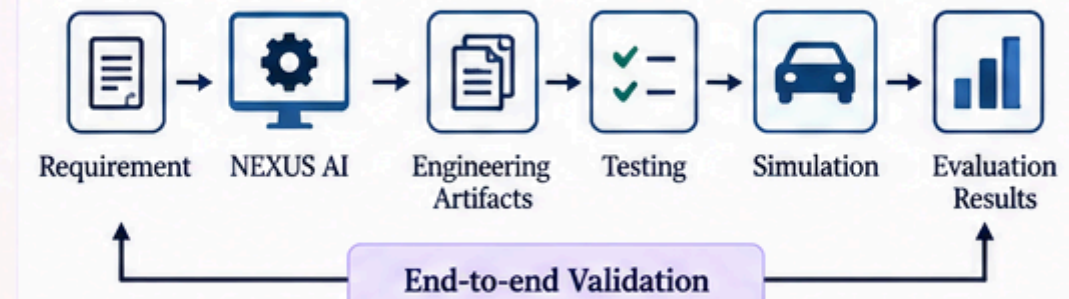


Key Goal



“To verify that the system does not only generate an answer, but also produces, tests and validates an engineering solution.”

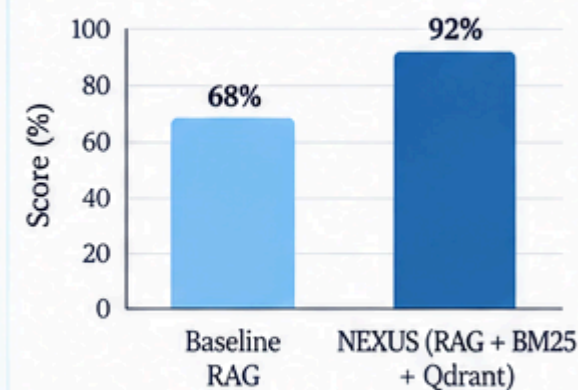
Complete Evaluation Flow



Results and Analysis

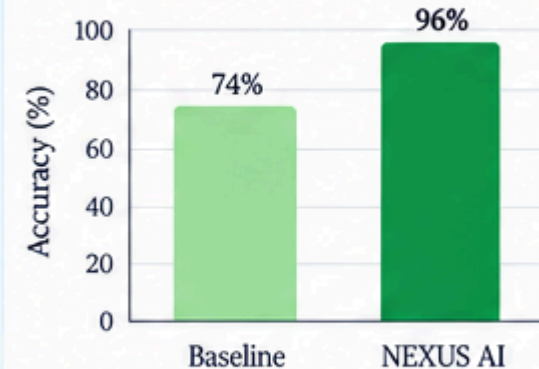
1 RAG Quality

Relevance of Retrieved Information



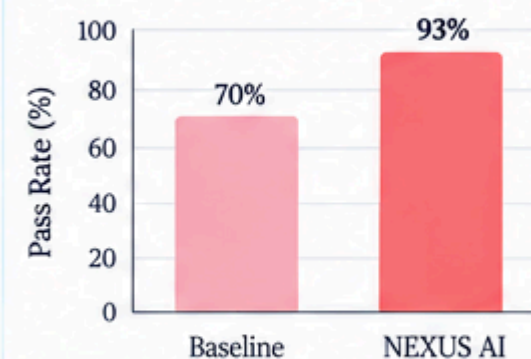
2 Citation Quality

Correctness of Source Citations



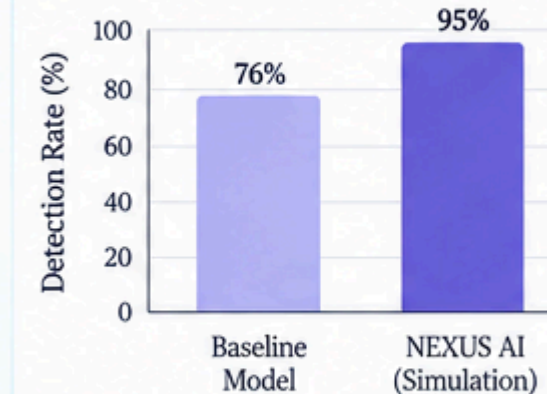
3 Code Quality

Generated Code Test Results



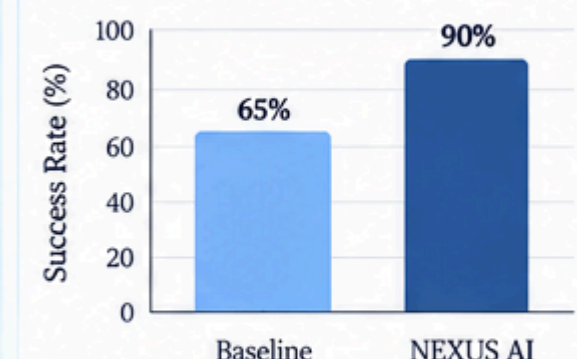
4 Simulation

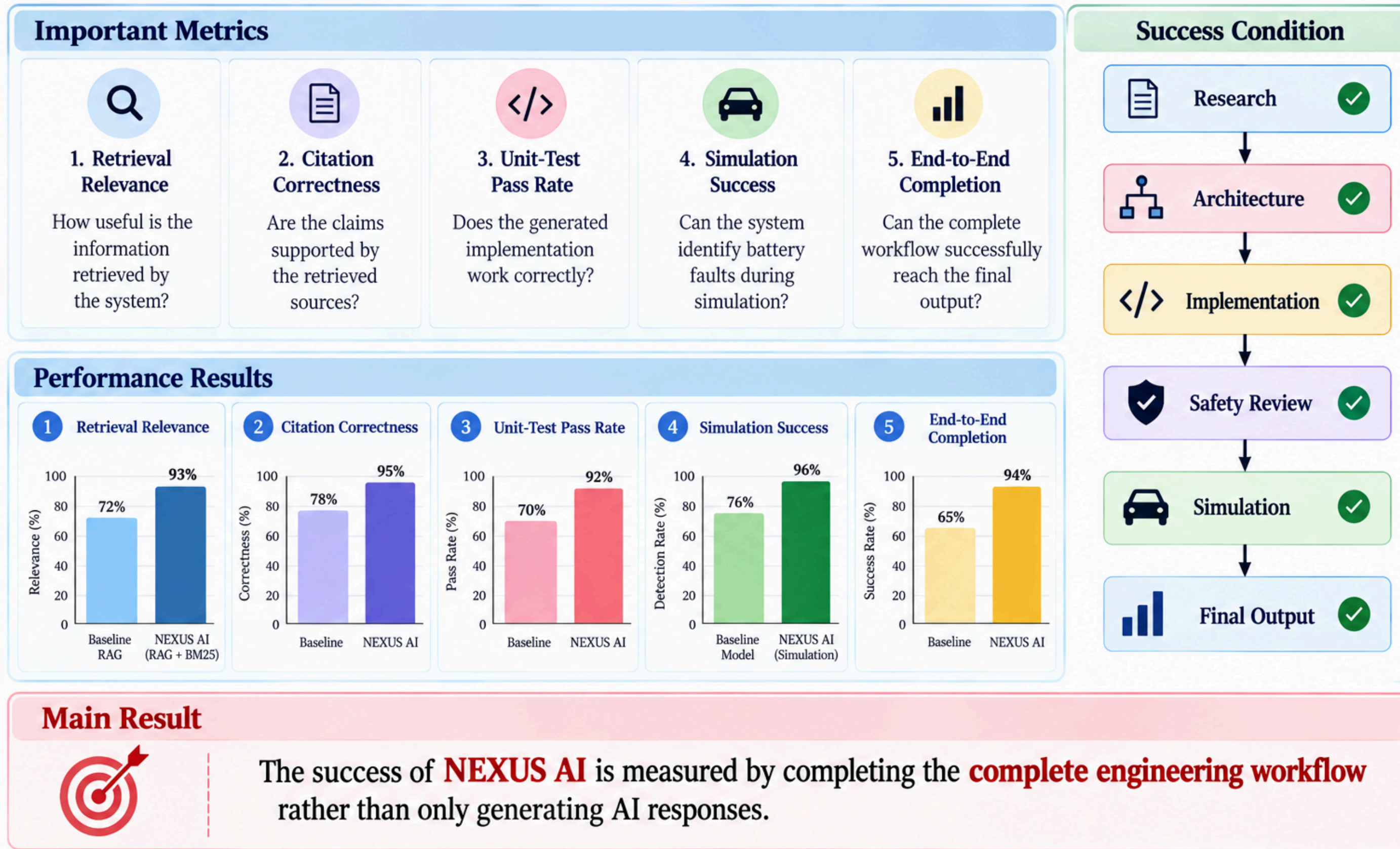
Fault Detection Performance



5 Workflow

End-to-End Success Rate





Feature	LLM Only	Basic RAG	NEXUS AI
Requirement Understanding	✓	✓	✓
Knowledge Retrieval	✗	✓	✓
Multiple AI Agents	✗	✗	✓
Code Generation	✓	✓	✓
Safety Review	✗	✗	✓
Simulation	✗	✗	✓
Reflection & Correction	✗	✗	✓
Human Approval	Limited	Limited	✓
Traceability	Limited	Limited	✓

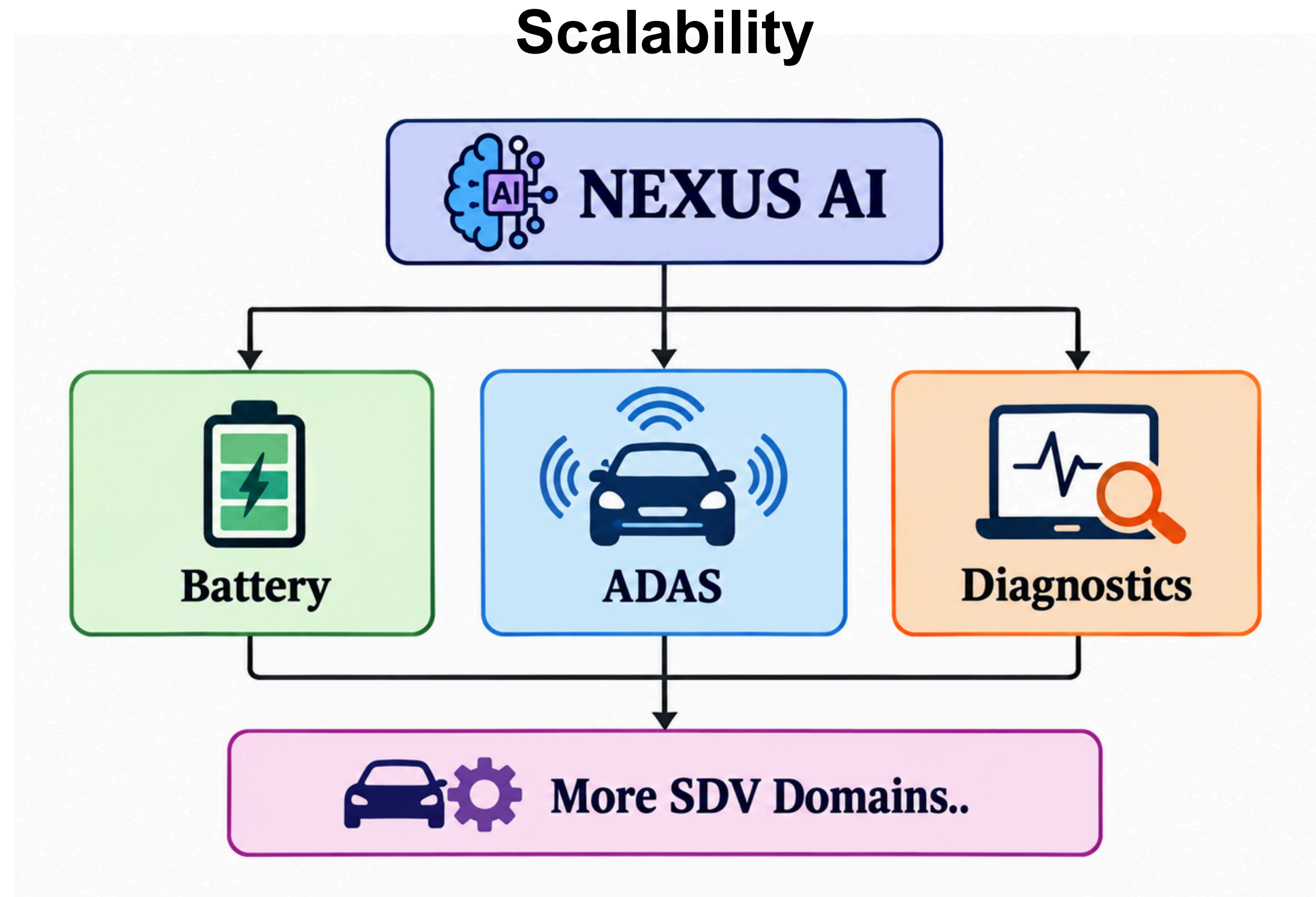
Our Innovation

NEXUS AI connects Research → Design → Implementation → Safety → Simulation → Correction in one continuous workflow.

Real-World Applicability

NEXUS AI can support:

- EV Battery Monitoring
- ADAS
- Vehicle Diagnostics
- Thermal Management
- Powertrain Systems
- Other SDV Applications



1. More automotive domains

Extend from battery monitoring to:

- ADAS
- powertrain
- thermal management
- diagnostics

2. More simulation scenarios

Increase:

- fault types
- vehicle scenarios
- environmental conditions

3. Improved evaluation

Conduct larger benchmark datasets and repeated experiments.

4. Advanced knowledge sources

Expand the curated knowledge base while respecting source licensing.

5. Production deployment

Introduce:

- authentication
- role-based access
- scalable model serving
- distributed simulation

6. Advanced HMI

connect PIXEL's machine-readable configuration to the React frontend.

NEXUS AI demonstrates how Agentic AI can move beyond conversational assistance toward an engineering workflow for Software-Defined Vehicles.

The system:

- understands requirements
- researches automotive knowledge
- designs architectures
- generates implementations
- performs engineering-oriented safety review
- validates through simulation
- detects failures
- performs reflection and correction
- involves humans for high-risk decisions
- generates HMI specifications
- maintains a complete engineering audit trail



Thank You

Department of Artificial Intelligence and Machine Learning